Name : Yashwin K Shetty

Roll number : 2611006200007

Apply the concepts of pipelining & parallel processing

CO3

1) Benefit of pipelines in processor

→ Benefits of pipeling in a processor

- Increases speed : Executes instructions faster
- Improves throughput : Multiple instructions are processed simultaneously at different stages
- Better CPU utilization : Makes efficient use of processor resources
- Reduces execution time : Improves overall processor performance
- Increases efficiency : Allows continous instruction processing

2) Difference b/w 3 stages & 5 stage Pipelines

→ 3 Stage vs 5 stage pipeline

Stages : 3 stage has fetch, decode, execute.
5 stage has Fetch, decode, execute, memory, write back

Clock speed : 3 Stage is lower because the execute stage does too much

5 stage is higher because each stage is shorter

Instruction in flight : 3 stage has up to 3.

5 stage has up to 5

Memory access : In 3 stage it is done inside execute, so load/store taken extra cycles

In 5-stage it has its own memory stage

Hazards : 3 stage has fewer & simpler ones

5 stage has more & needs forwarding & hazard detection

Hardware & power : 3 stage is simpler & lower

5 stage is more complex & higher

7. Example : 3 stage in Arm 7
5 stage in Arm 9 or MIPS

3) Explain the relationship of capacitance, frequency & power dissipation in single core to multicore transition problem make it short

-> Core formula : Dynamic power $P = C \cdot V^2 \cdot f$

C (capacitance) -> The total switching capacitance of the chip. More transistors or cores means more C

f (frequency) -> How often the circuit switches. Power rises linearly with f

V (voltage) -> needed to support higher f & Since P depends on V^2, power grows roughly as f^3 when voltage is scaled up with freq

Single core problems : To get more performance you raise f, which needs higher V. power & heat shoot up, so frequency scaling hit a limit

Multicore solution : Instead of one fast core use several slower cores

· Each core runs at a lower f & lower V, so power per core drops sharply

· Total C increases but the savings from lower V & f outweigh it

· performance is gained through parallelism rather than clk speed

4) challenges of systems with customer expecta-tions in single & multicore systems make it short

→ challenges of systems vs customer expectations

1. Single core systems

· Slower execution of multiple tasks

· limited multitasking performance

· longer response time

· Difficulty handling heavy workloads

2. Multi core systems

· Complex task scheduling

· Higher power consumption

· Increased system cost

- Difficulty in balancing workloads across cores

5) what do you mean by hazards explain types of hazards with example?

-> Hazards are problems that occur in a computer processor when instructions cannot execute properly in the expected order. They may cause delays or incorrect results during instruction execution in a pipeline

Types of hazards

Mainly three types of hazards

Structural hazard

occurs when two instructions need the same hardware resource at the same time

Ex: Two instructions try to access the same memory simultaneously

Data Hazard

occurs when one instruction depends on the result of a previous instruction that

han not yet completed

Control hazard

Occurs when the processor fetches a branch or jump instruction & does not know which instruction to execute next

The processor must decide which instruction to execute next